ETH Zurich
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FTS-Diffusion: Generative Learning for Financial Time Series

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Machine Learning in Finance and Complex Systems

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Outline

Outline

The problem: financial time series are hard to generate

Empirical stylized facts

- Heavy-tailed returns; rare but consequential crises.
- Volatility clustering, long memory in $|r_t|$.
- Scale-invariant patterns at irregular timescales.
- Single realised history (no resampling).

Why standard generators fail

Fixed-window models segment in **calendar time**, not **pattern time**. The same shape at twice the duration becomes a different sample.

The paper's contribution

FTS-Diffusion (Yu et al., ICLR 2024): a three-module pipeline.

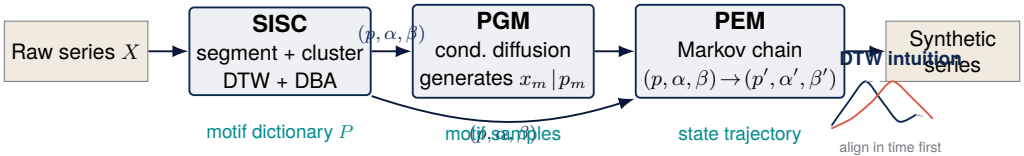
- **SISC**: Scale-Invariant Subsequence Clustering, discovers a finite dictionary of variable-length motifs using DTW.
- **PGM**: Pattern Generation Module, a conditional diffusion network that synthesises new latent motifs.
- **PEM**: Pattern Evolution Module, a Markov chain over latent states (p_m, α_m, β_m) steering long-horizon dynamics.

Our goal

Replicate the pipeline, stress-test it on three assets (S&P 500, GOOG, ZC=F), and isolate the methodological choices that matter.

Outline

FTS-Diffusion pipeline



State variables

- p: pattern identity (cluster label).
- α: duration scale factor (length stretch).
- β: magnitude scale factor (amplitude).

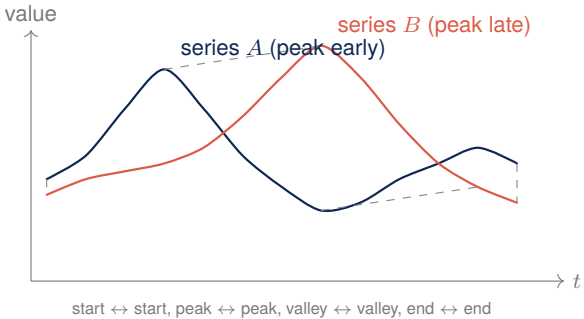
DTW: align in time before measuring distance.
Two segments stay close even if one is stretched.

Why K-means++ for centroid init?

- K-means++ seeds far-apart centroids, more stable clusters.
- Plain K-means can collapse two centroids onto the same regime.

Outline

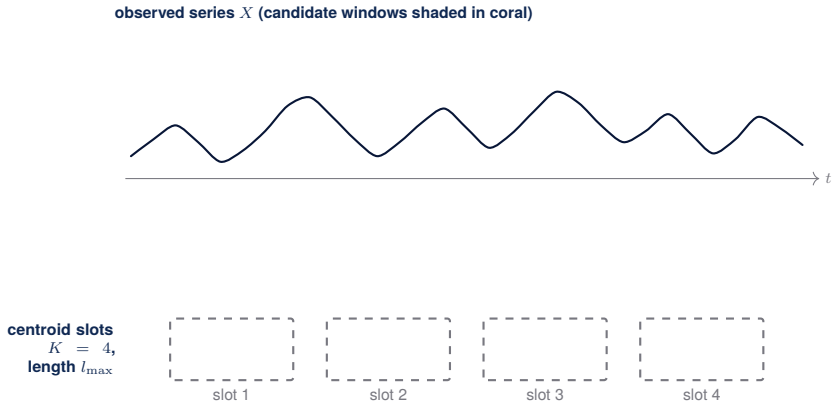
DTW: align in time before measuring distance



DTW finds the best monotone alignment of the two series, then measures distance along that alignment instead of pointwise.

- Pointwise distance would say *"these are very different"* (peaks misaligned in time).
- DTW distance says *"same shape, just stretched in time"* → small.
- This is exactly what we need for financial motifs that recur with different durations.

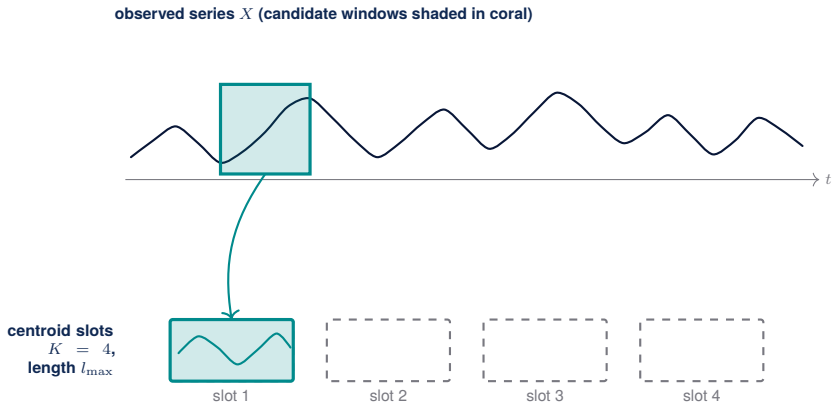
SISC, Step 1: Initializing centroids (K-means++ for DTW)



Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \;\propto\; \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.

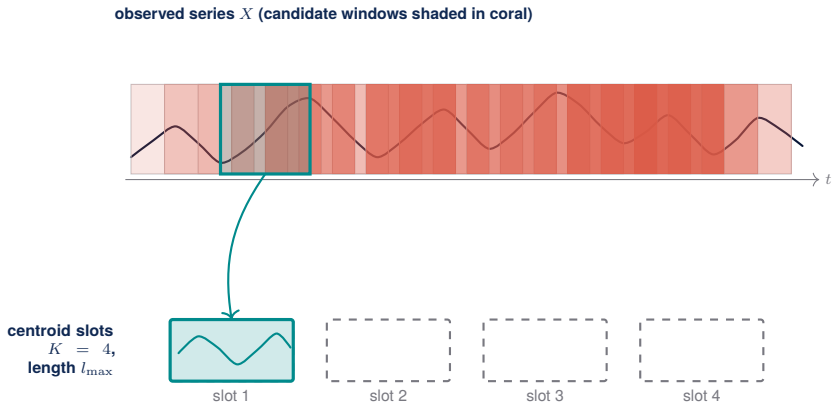
Step 0: $K = 4$ centroids of fixed length $l_{\max}$ to be chosen.

SISC, Step 1: Initializing centroids (K-means++ for DTW)



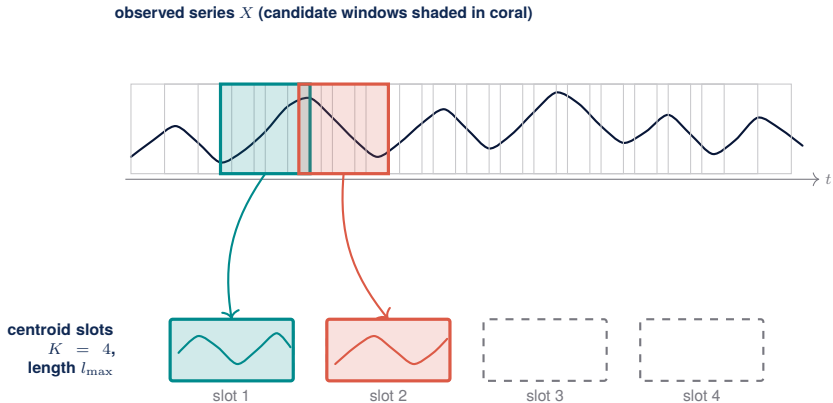
Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \propto \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.
Step 1: p_1 sampled uniformly from all length-$l_{\max}$ windows.

SISC, Step 1: Initializing centroids (K-means++ for DTW)



Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \propto \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.
Step 2: candidate windows weighted by distance to p_1 (coral = far).

SISC, Step 1: Initializing centroids (K-means++ for DTW)



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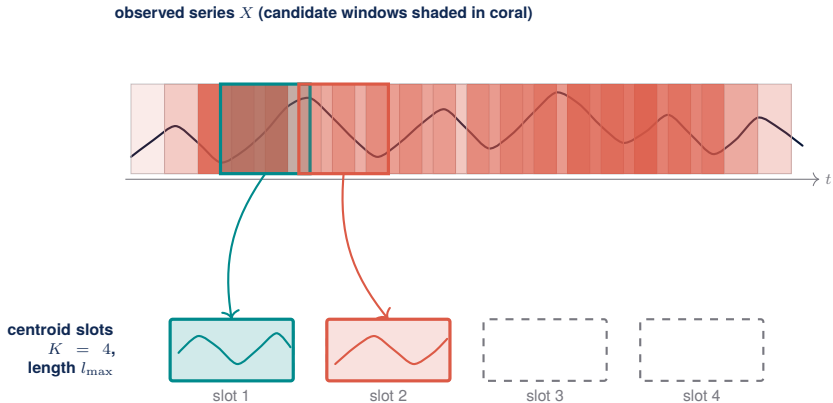
Step 3: p_2 sampled with prob. $\propto \widetilde{\mathrm{DTW}}(\cdot, p_1)$.

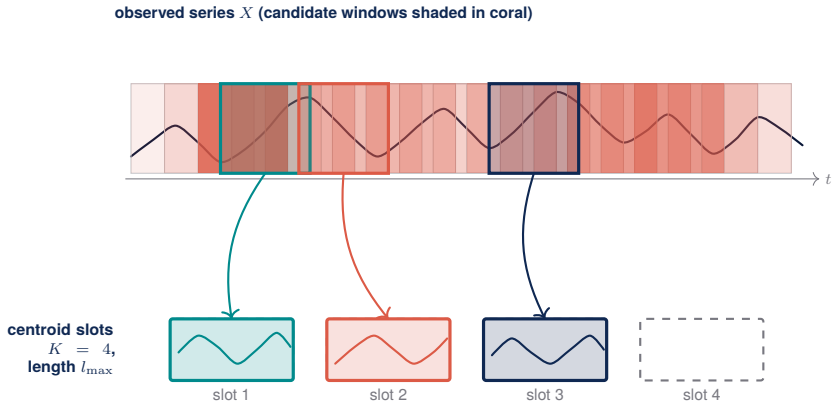
SISC, Step 1: Initializing centroids (K-means++ for DTW)



Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \;\propto\; \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.

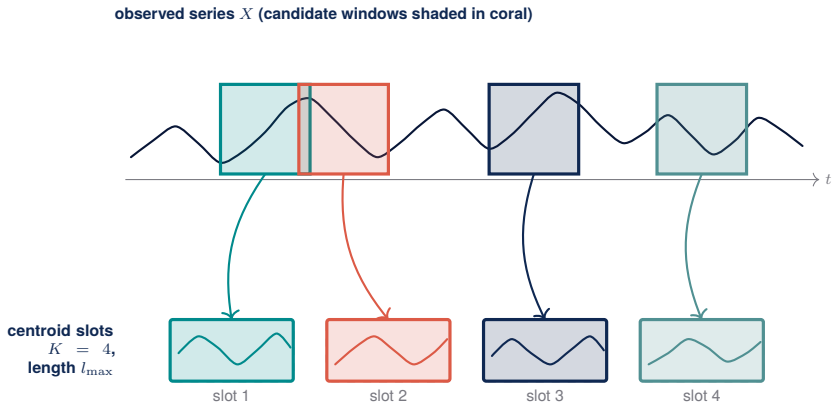
Step 4: recompute weights against $\{p_1, p_2\}$.

SISC, Step 1: Initializing centroids (K-means++ for DTW)



Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \propto \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.
Step 5: p_3 sampled with prob. $\propto \min_{k' \le 2} \mathrm{DTW}(\cdot, p_{k'})$.

SISC, Step 1: Initializing centroids (K-means++ for DTW)



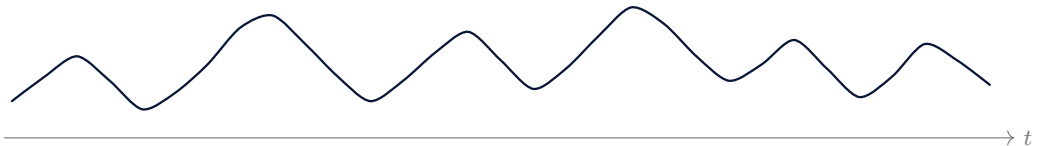
Sampling rule: $\Pr\big(p_{k+1} = X[t : t + l_{\max}]\big) \propto \min_{k' \le k} \mathrm{DTW}\big(X[t : t + l_{\max}],\, p_{k'}\big)$.

Step 6: p_4 chosen; initialisation complete.

SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids

$P = \{p_1, \ldots, p_4\}$



SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids
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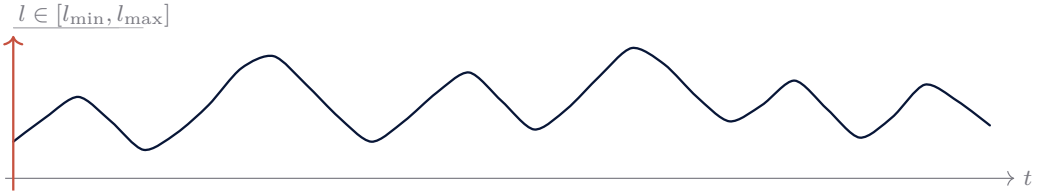


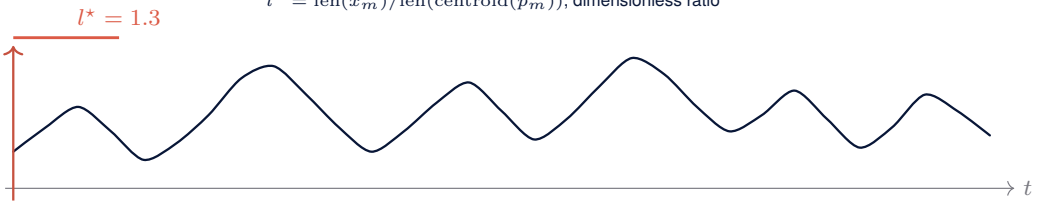
$K \cdot (l_{\max} - l_{\min} + 1) = 168$ DTW evals per step

SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids

$P = \{p_1, \ldots, p_4\}$

$l^\star = \text{len}(x_m)/\text{len}(\text{centroid}(p_m))$, dimensionless ratio



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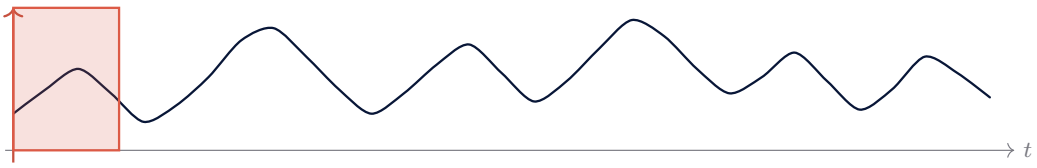

greedy: emit best $(l^\star, k^\star)$, advance

SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids
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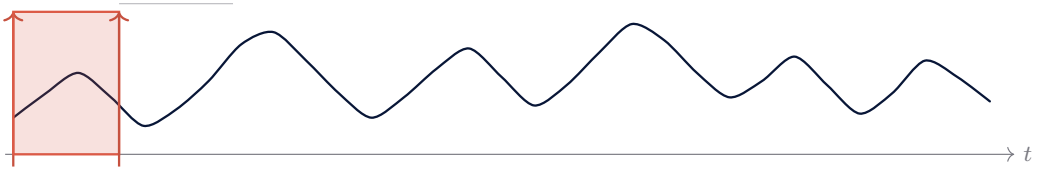

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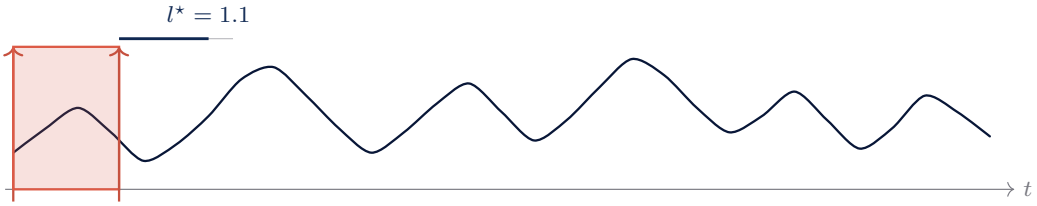

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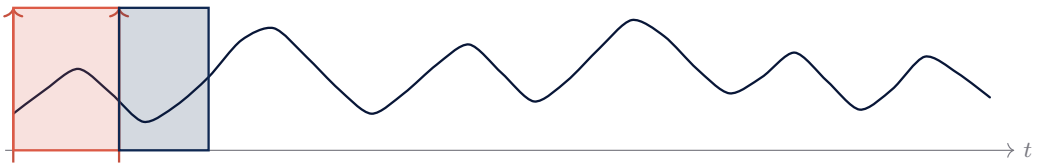

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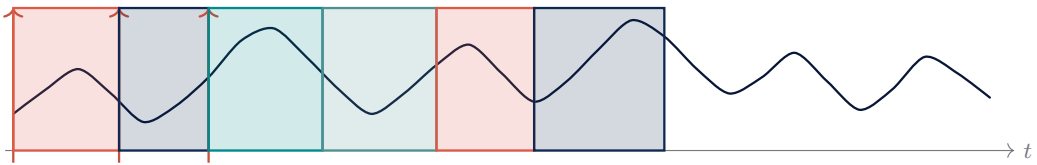
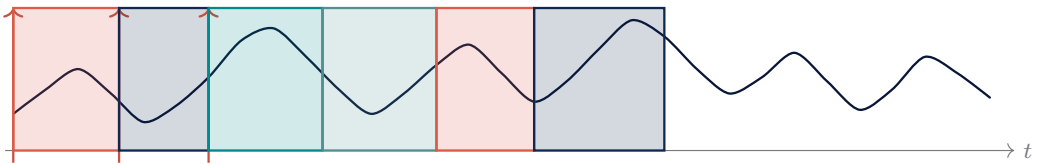


variable-length segmentation continues

SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids
$P = \{p_1, \ldots, p_4\}$

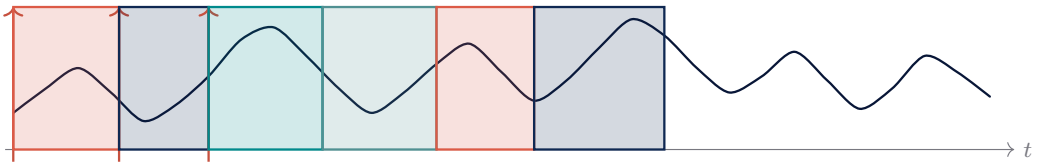


variable-length segmentation continues

cluster C_k (members)

SISC, Step 2: Greedy segmentation + DBA centroid update

current centroids
$P = \{p_1, \ldots, p_4\}$



SISC, Step 2: Greedy segmentation + DBA centroid update

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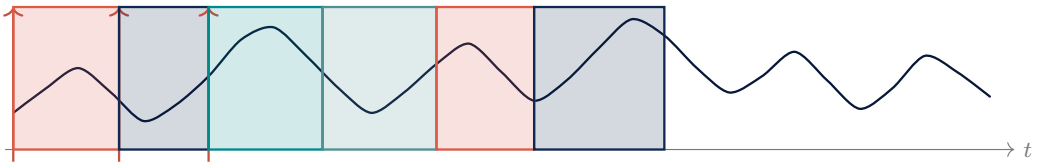

SISC, Step 2: Greedy segmentation + DBA centroid update

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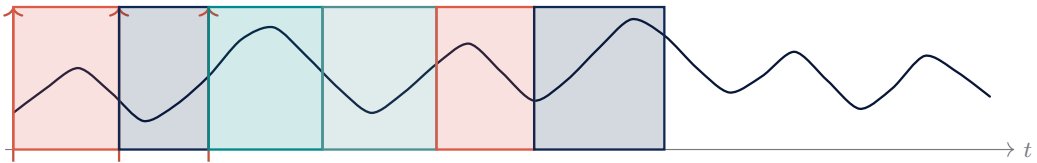
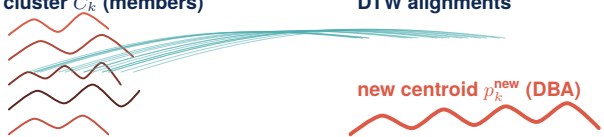


variable-length segmentation continues

cluster C_k (members)

DTW alignments

new centroid p_k^{new} (DBA)



DBA (DTW Barycenter Averaging): centroid is the **average over aligned values**, not pointwise. It produces a barycenter of variable-length sequences in DTW space.

$$p_k^{\text{new}} = \arg\min_{p} \sum_{x \in C_k} \mathrm{DTW}(x, p)^2.$$

SISC, Step 3: Iterate until convergence

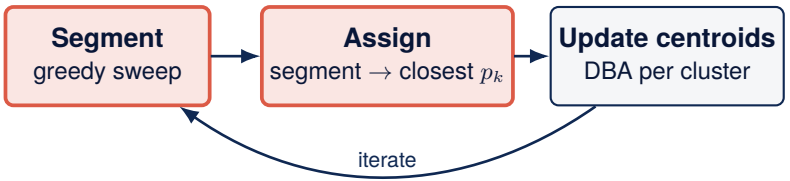
iterations left

20

one tick = one full cycle (segment+assign) + update

Step 0: 20-iteration budget, no work yet.

SISC, Step 3: Iterate until convergence



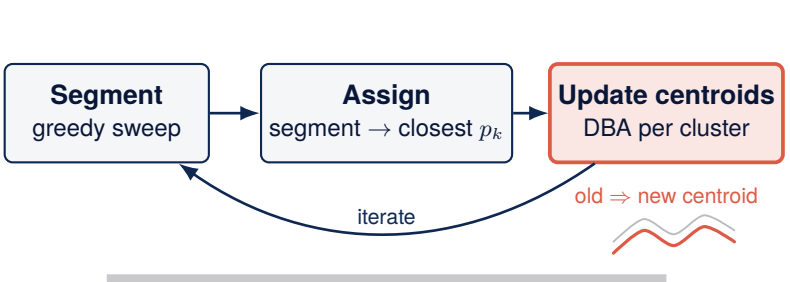
iterations left

20

one tick = one full cycle
(segment+assign) + update

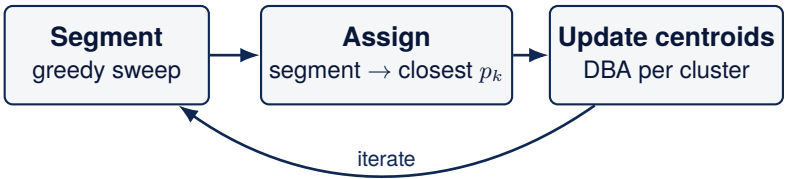
Iteration 1, phase: *segment* + *assign* (one conceptual step).

SISC, Step 3: Iterate until convergence



Iteration 1, phase: *centroid update* (DBA). Counter still 20.

SISC, Step 3: Iterate until convergence



iterations left

19

one tick = one full cycle (segment+assign) + update

End of iteration 1: full cycle complete, budget ticks to 19.

SISC, Step 3: Iterate until convergence

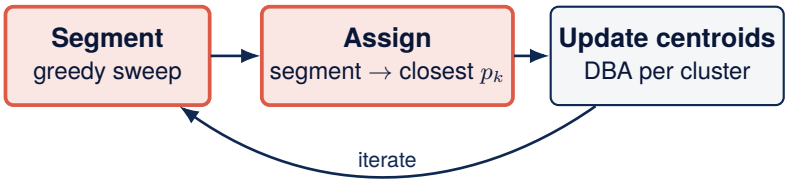
iterations left

12

one tick = one full cycle (segment+assign) + update

Fast-forward several iterations: shifts shrink, budget reaches 12.

SISC, Step 3: Iterate until convergence



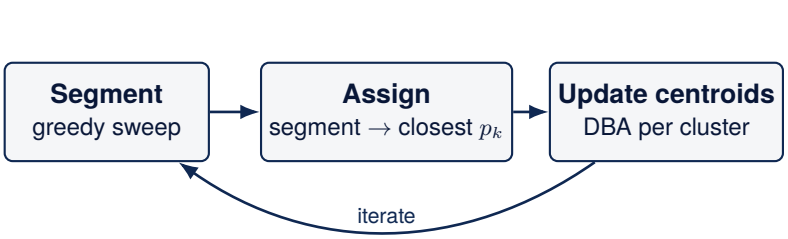
iterations left

8

one tick = one full cycle (segment+assign) + update

Stability check: are assignments unchanged across cycles?

SISC, Step 3: Iterate until convergence



iterations left

8

one tick = one full cycle (segment+assign) + update

Convergence. Stops on stable assignments OR budget exhausted.

Limits of the SISC algorithm

Local minima risk

SISC is a non-convex optimisation: greedy segmentation followed by iterative centroid update. It can stall when motifs are noisy or weakly separated.

Initialization sensitivity

Bad initial centroids yield a bad final segmentation. This is exactly why we use K-means++ for the seeding step (back-reference to the pipeline slide).

Hyperparameter sensitivity

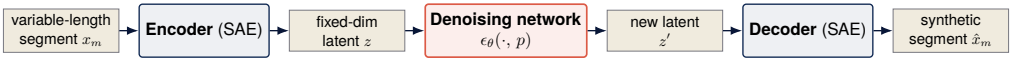
K and $[l_{\min}, l_{\max}]$ must be chosen up front. Mis-set values either over-fragment (too many tiny motifs) or under-segment (one motif swallows the series).

Inherent to any non-convex clustering procedure on time-series data: we don't take a clean segmentation for granted.

Outline

Step 2: pattern generation

How does diffusion know what to generate? Conditioning happens in latent space.



Conditional forward step (paper §4.2):

$$q(x^i \mid x^{i-1}) = \mathcal{N}\Big(x^i;\ \sqrt{1-\beta}\,(x^{i-1} - p),\ \beta I\Big).$$

Noise is added to the residual $(z - p)$, so the denoiser learns to produce segments resembling the centroid of pattern p, not arbitrary signals.

- SAE encoder maps variable-length → fixed-dim latent.
- Diffusion runs in the latent, conditioned on p.
- SAE decoder maps the new latent back to a segment.
- **Coherent pattern evolution** prevents learning pure noise.

Scaling autoencoder

From variable-length segments to fixed latent representations

Encoder

Maps an observed variable-length segment to a fixed latent:

$$x_m \longrightarrow x'_m.$$

Decoder

Reconstructs a financial segment from the latent:

$$x'_m \longrightarrow \hat{x}_m.$$

- Handles heterogeneous segment lengths.
- Makes diffusion feasible in a fixed-dimensional latent space.
- Implemented with LSTM / GRU layers.

Pattern-conditioned diffusion network

Learning to denoise latent pattern representations

Forward diffusion

Gradually corrupt the latent with Gaussian noise:

$$q(x^i \mid x^{i-1}) = \mathcal{N}\big(x^i;\ \sqrt{1-\beta}(x^{i-1} - p),\ \beta I\big).$$

Reverse denoising

A network learns to recover the previous step, conditional on pattern p:

$$p_\theta(x^{i-1} \mid x^i) = \mathcal{N}\big(x^{i-1};\ \mu_\theta(x^i, i, p),\ \beta I\big).$$

Intuition

Map noise to realistic pattern-level market segments.

Training loss: three losses, asymmetric weights

Concrete weights from `pgm_params` (`model_params.py:32-53`)

Composite objective

$$\mathcal{L}_{\text{total}} = \underbrace{\lambda_{\text{SAE}}}_{=0.98}\mathcal{L}_{\text{recon}} + \underbrace{\lambda_{\text{diff}}}_{=0.01}\mathcal{L}_{\text{diff}} + \underbrace{\lambda_{\text{scale}}}_{=0.01}\mathcal{L}_{\text{SoftDTW}}.$$

`pad_weight`$= 1$ (padding weighted equally with data) | `reduction='sum'` (MSE summed, not averaged).

Commentary

- Three losses summed: SAE reconstruction (price space), PCDM diffusion (latent noise residual), SoftDTW(x, z) (latent interpretability).
- The $0.98 : 0.01$ ratio is **98:1**. The gradient is tilted toward reconstruction; diffusion receives $< 1\%$ of the signal, by design.
- `mse(reduction='sum')` + `pad_weight`$= 1$: the SAE spends a large fraction of its gradient budget on padding zeros, which directly caps how much the diffusion can learn.

Open question

Would raising λ_{diff} (e.g. 0.1) and zeroing `pad_weight` improve downstream MAPE? *Hypothesis:* an un-suppressed diffusion produces sharper latent centroids, hence more stable MC transitions, the weak link the PEM feeds into TATR.

Outline

Step 3: pattern evolution

Markets evolve through regimes; PEM learns the empirical transition kernel

Markets evolve through regimes (bull/bear, low-vol/high-vol). The PEM learns the empirical transition kernel between the latent regimes that SISC discovered.

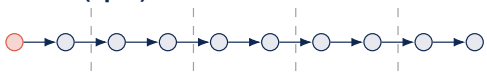
What the PEM does

1. Governs the **temporal structure** of the synthetic series.
2. Assumes Markov dynamics: $S_{m+1} \sim Q(\cdot \mid S_m)$.
3. **Sensitive to the rollout protocol** (headline of the next slides).
4. Iterated **autoregressively** at sampling time: MAPE worsens as more synthetic data is fed in, errors compound.
5. **Low-dimensional**: collapses each segment to (p_m, α_m, β_m).

authors (reset)

[reset] → chain → [reset] → chain . . .

ours (split)



[start] → continuous chain → sliced into blocks

Authors **reset** the chain at the start of every downstream block. We let the chain **evolve continuously** and slice it. This subtle choice flips the result for S&P 500 (we'll see how in the results section).

Pattern evolution network

Classification for patterns, regression for scales

Transition network

$$(\hat{p}_{m+1}, \hat{\alpha}_{m+1}, \hat{\beta}_{m+1}) = \phi(p_m, \alpha_m, \beta_m).$$

Outputs

- next pattern p_{m+1};
- next duration scale α_{m+1};
- next magnitude scale β_{m+1}.

Learning tasks

- pattern: classification;
- duration: regression;
- magnitude: regression.

$$\mathcal{L}(\phi) = \mathbb{E}_{x_m}\Big[\ell_{CE}(p_{m+1}, \hat{p}_{m+1}) + \|\alpha_{m+1} - \hat{\alpha}_{m+1}\|_2^2 + \|\beta_{m+1} - \hat{\beta}_{m+1}\|_2^2\Big].$$

Full synthetic time-series generation

Putting the three modules together

1. Start from an initial historical segment.
2. Use the evolution network to predict $(\hat{p}_{m+1}, \hat{\alpha}_{m+1}, \hat{\beta}_{m+1})$.
3. Generate the next segment via the pattern-conditioned diffusion model.
4. Decode it into a variable-length financial segment.
5. Append the segment to the synthetic series.
6. Repeat until the desired path length is reached.

Intuition

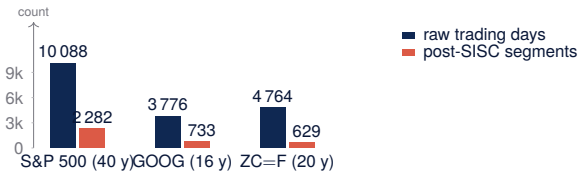
Local motifs are produced by diffusion; global temporal structure is controlled by the evolution network.

Outline

Training data per asset

Strong size asymmetry across the three assets

Asset	Type	Period	Years (days)	Post-SISC seg. (K)
S&P 500	index	1980–2020	40 (10 088)	2 282 (14)
GOOG	stock	2004–2020	16 (3 776)	733 (11)
ZC=F	future	2000–2020	20 (4 764)	629 (11)



Data pipeline

- **Source:** Yahoo Finance daily close.
- **Modality:** **raw prices**, not returns.
- **FTS-Diffusion split:** 80 % train / 20 % test.
- **Downstream LSTM:** carved from the 20 % test slice.

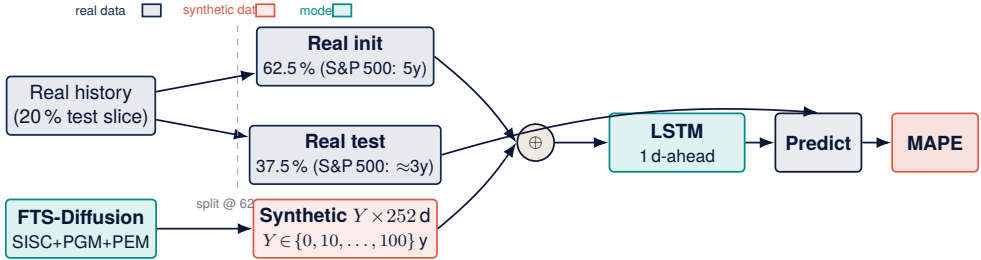
Size asymmetry

After segmentation, S&P 500 is $\approx 3\times$ GOOG and $\approx 3.6\times$ ZC=F.

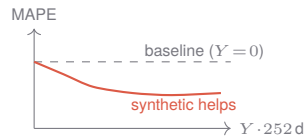
Outline

TATR: Train on Augmentation, Test on Real

Does FTS-Diffusion synthetic data *help* a forecaster?



- **Train:** `real_init` $\oplus$ `synth`$[0 : Y \cdot 252]$.
- **Test:** real held-out (never seen during training).
- **Sweep:** $Y \in \{0, 10, 20, \ldots, 100\}$ years of synthetic.
- **Score:** MAPE on real test, mean over 15 seeds.



One question only

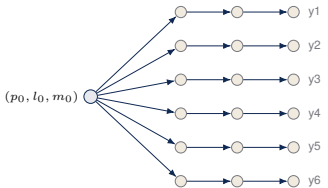
Does adding Y years of synthetic data **reduce** the LSTM's MAPE on real held-out data? A curve below the $Y = 0$ baseline says *yes*. 5y/3y are S&P 500; the same 62.5/37.5 ratio is kept on GOOG and ZC=F.

Outline

Three ways to produce the $Y \cdot 252$ synthetic days

Same FTS-Diffusion, three Markov-chain rollouts

authors



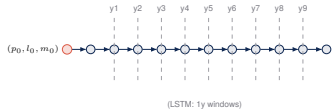
· · · (10 chains total)

Re-init every year from the same (p_0, l_0, m_0).

Pro: paper-faithful.

Con: no cross-year MC.

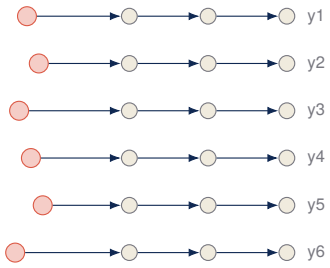
split



Single continuous MC; year cuts only affect the LSTM rolling window.

Same budget; MC persists across years.

random_init



· · · (10 chains total)

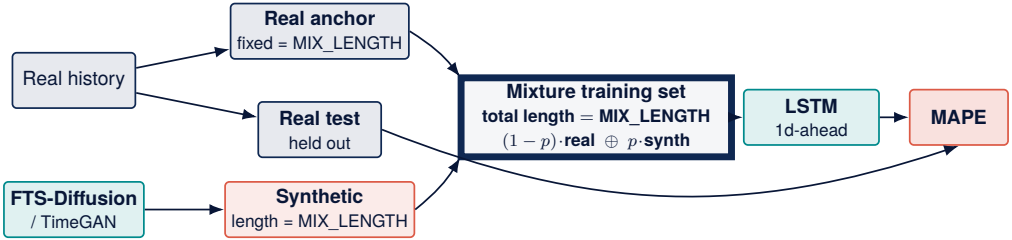
Re-init each year; init state $\sim$ `SEGMENTS_INIT`.

Diagnostic: isolates init from MC.

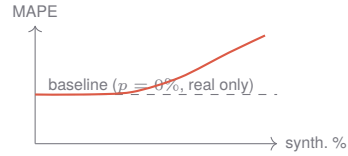
Outline

TMTR: Train on Mixture, Test on Real

Fixed-budget composition sweep; the LSTM trains on a mixture of real and synthetic



- Total training length is constant (S&P 500: 5y). Only the *composition* varies.
- Sweep: synthetic proportion $p \in \{0\%, 10\%, \ldots, 100\%\}$.
- Score: MAPE on the real held-out test set, averaged over 20 seeds.
- **Difference vs TATR:** TATR grows the training set with synthetic. TMTR keeps total size fixed and **replaces** a fraction with synthetic.

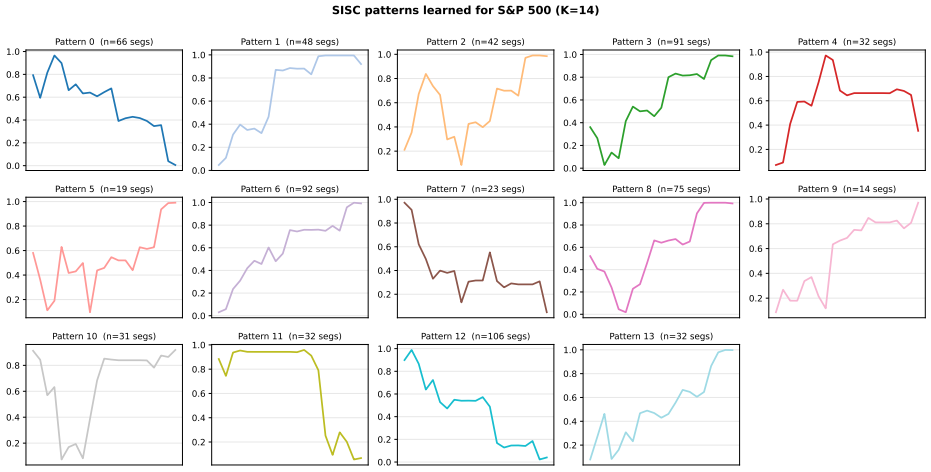


What TMTR asks

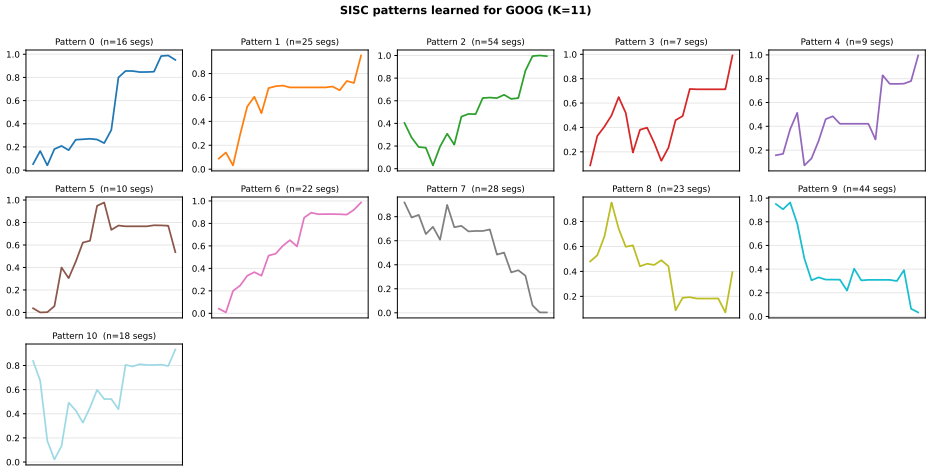
At what synthetic proportion does the LSTM stop matching the real-only baseline? A flat curve $\Rightarrow$ the

Outline

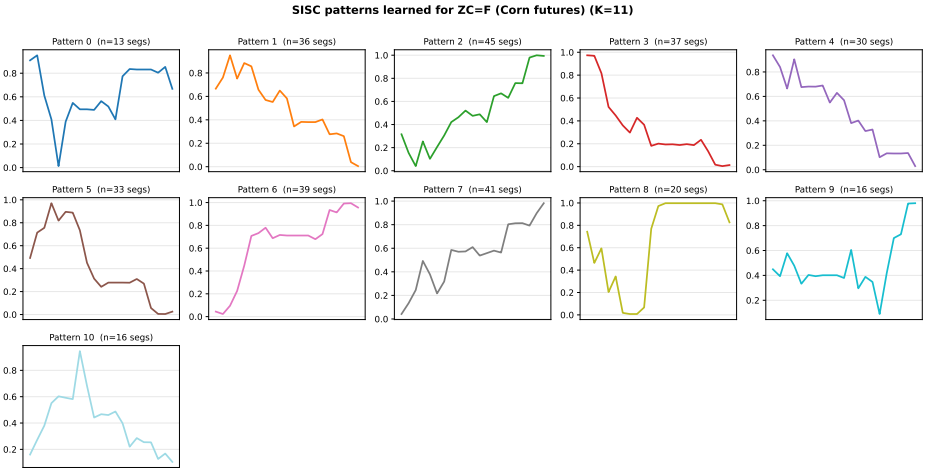
S&P 500: learned motifs ($K = 14$)



GOOG: learned motifs ($K = 11$)



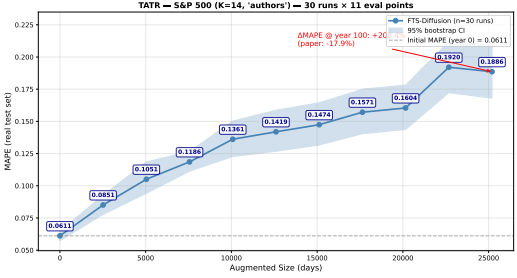
ZC=F (corn futures): learned motifs ($K = 11$)



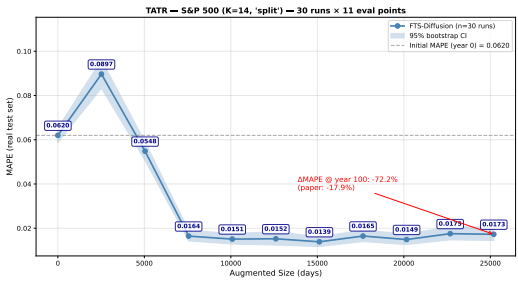
Outline

S&P 500, $K = 14$: authors vs split

authors



split



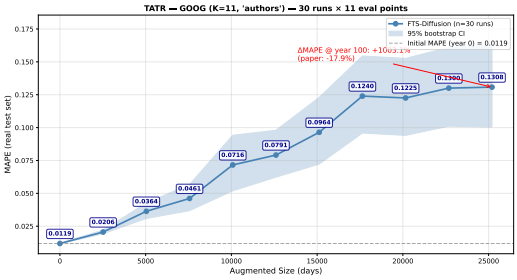
MAPE delta: $+208.47\%$ vs. baseline.

Split: continuous chain across years.

Same data budget; only the MC connectivity changes between protocols.

GOOG, $K = 11$: authors vs split

authors



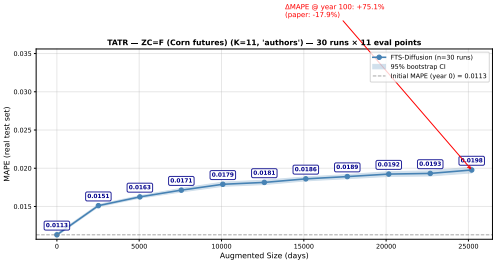
split (TODO: insert split panel, figure pending)

per-run CSVs available
under `results/tatr/goog/k11/split/`;
final plot not yet generated
(figure pending)

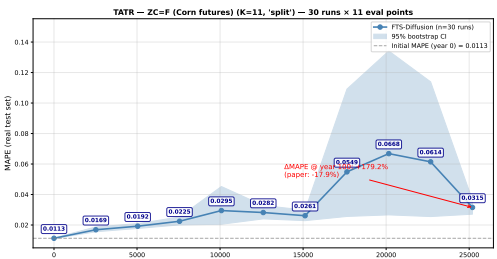
$K = 11$ on GOOG: smaller post-SISC budget shrinks the cluster count proportionally.

ZC=F, $K = 11$: authors vs split

authors

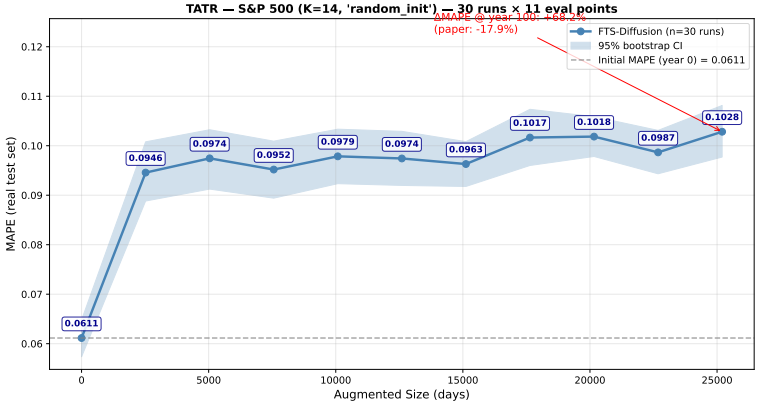


split



Same $K = 11$ as GOOG; the MAPE level differs because the motif library does.

S&P 500, $K = 14$: diagnostic, random init

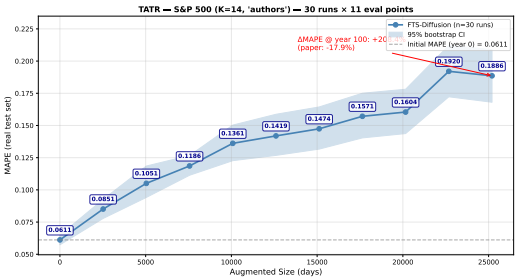


Random init isolates the initialisation effect from the MC-evolution effect.

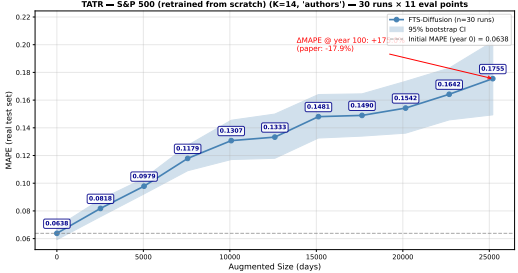
Authors' checkpoint vs. trained-from-scratch (S&P 500, $K = 14$)

Same authors protocol; only the PGM+PEM weights differ

authors' checkpoint



trained from scratch



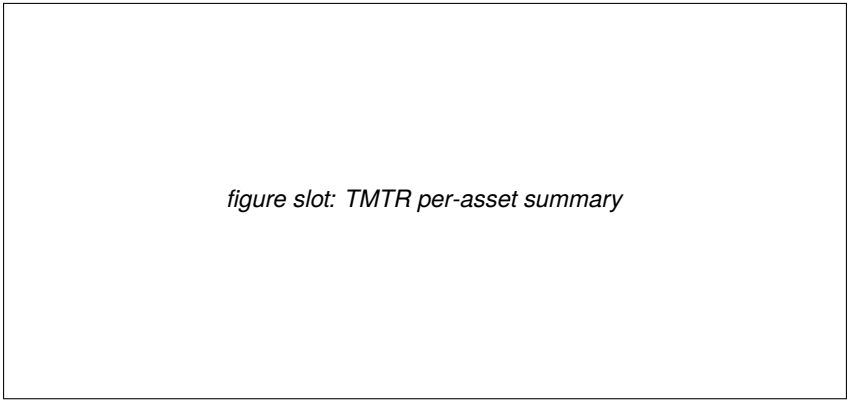
Why $K = 14$ here: keeps the authors' data budget on S&P 500; on the smaller assets we use $K = 11$ because the post-SISC training set shrinks proportionally.

Outline

TMTR: Train More, Test Real

Status

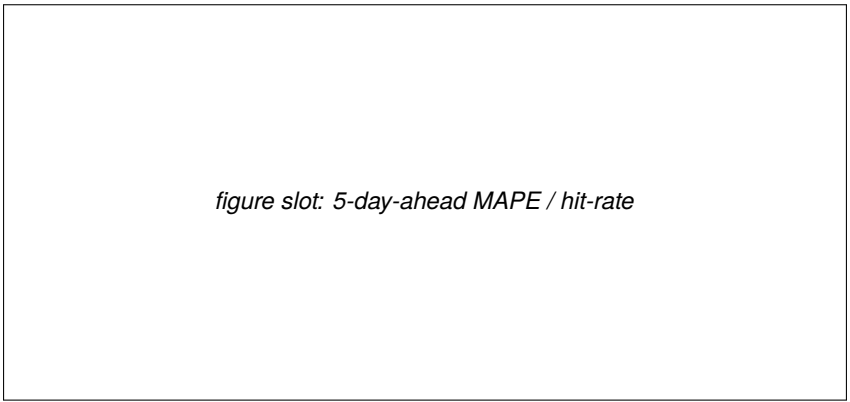
Experiment in progress. Figures will be inserted here once the runs complete.



5-day-ahead forecasting

Status

Experiment in progress. No numerical results to report yet.



Outline

Reproducibility issues uncovered

- **Hard-coded LSTM warm-up.** `init_period = 252*5` in the downstream LSTM assumes 5 years of trading days. Breaks silently for GOOG and ZC=F (insufficient history); both run but evaluate the LSTM in its warm-up regime.
- **Device mismatch in `train_evolution_module`.** Target tensors are not moved to GPU. Forward stays on CPU even with a CUDA model; training is 8–$10\times$ slower and the optimiser trajectory differs.
- **x-range multiplier in `plot_downstream_tatr`.** Augmentation axis multiplied by 10 in plotting; mis-labels "$10\times$" as "$100\times$".
- **Under-specified synthetic-data protocol** — the paper does not pin down "authors" vs. continuous-chain rollout (split).

Outline

Why exact reproduction was difficult

- **Synthetic-data protocol is under-specified.** The paper does not fully specify whether synthetic paths are generated as independent segments, one continuous chain, or a continuous chain later split into evaluation blocks.
- **Downstream split is not portable across assets.** The released downstream LSTM uses `init_period = 252*5`. This is compatible with the S&P 500 setup, but GOOG and ZC=F do not have enough held-out history under the same split. Any valid GOOG/ZC=F run therefore requires an adapted protocol.
- **GPU training path requires a patch.** In `train_evolution_module`, target tensors are not moved to the model device. On CUDA this creates a device mismatch, so the intended GPU path is not directly reproducible without modifying the code.
- **Published trends are sensitive to protocol choices.** Our experiments show that changing the synthetic rollout protocol can materially change the TATR trend, even with the same assets and similar hyperparameters.

Markov-chain stationarity

Setup

PEM defines a Markov chain over $s_m = (p_m, \alpha_m, \beta_m)$. Stationarity would require a fixed invariant π with $\pi^\top T = \pi^\top$.

- Empirical T (per asset, per K) is right-stochastic; principal left-eigenvector gives π.
- Stationarity check: simulate long chains, monitor $\|\hat{\pi}_t - \pi\|_1$ over time.
- Drift check: compare mean motif occupancy in real vs. synthetic across the chain.

Open question — empirical burn-in

Under the `split` protocol, no clear stationarity is reached within one simulated year. How many segments of burn-in are needed before the chain forgets its init state? Not answered here.

Outline

Limitation 1: motif discovery is fragile

The whole generator depends on the first segmentation step

Discrete motif extraction

SISC turns a continuous time series into a finite dictionary of motifs. This is powerful, but it creates several sensitivities.

- The number of clusters K and length range $[l_{\min}, l_{\max}]$ are important hyperparameters.
- Greedy segmentation can misplace boundaries between adjacent motifs.
- DTW captures shape similarity, but may ignore economically meaningful differences.
- Motifs are learned offline; new market regimes may require re-clustering.

Implication: generation quality is only as good as the learned motif vocabulary.

Limitation 2: local patterns may miss long-range structure

Short motifs are not the same as full market dynamics

Captured well

- Local shape recurrence
- Heavy-tailed return behavior
- Volatility clustering proxies
- Segment-level scaling

Harder to capture

- Long-memory dependence
- Regime persistence
- Macro or news conditioning
- Rare crises and structural breaks
- Cross-asset co-movement

The learned transition is essentially a compressed state evolution:

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}).$$

Limitation 3: realism does not imply no artificial alpha

Stylized facts are necessary but not sufficient

The paper evaluates whether synthetic series resemble real data through distributional tests, stylized facts, and downstream prediction experiments.

Remaining concern

A synthetic generator can match heavy tails and volatility clustering, while still introducing **systematic predictable drift** under a simple trading signal.

- Good marginal fit does not guarantee correct conditional dynamics.
- Prediction experiments may depend on the model architecture used for evaluation.
- Leakage is possible if motif extraction or scaling is fitted using future data.
- A generator useful for augmentation should not create spurious predictability.

Why no predictable drift matters

Martingale intuition for financial returns

A standard benchmark assumption in financial market modelling is that returns should not contain an easily exploitable conditional mean under the available information:

$$\mathrm{E}[r_{t+1} \mid \mathcal{F}_t] = 0,$$

or, equivalently, for every admissible test function $g(\mathcal{F}_t)$,

$$\mathrm{E}[r_{t+1} g(\mathcal{F}_t)] = 0 = (\mathrm{Cov}(r_{t+1}, g(\mathcal{F}_t)) + \mathrm{E}[r_{t+1}]\,\mathrm{E}[g(\mathcal{F}_t)]) \,..$$

Financial interpretation

Given the information available at time t, the next return should not be systematically predictable. Otherwise, the generated data may contain **artificial alpha**.

- Real financial returns are noisy and close to unpredictable at short horizons.
- A good synthetic generator should match stylized facts without creating fake trading signals.
- Therefore, distributional realism alone is not enough: we also need a conditional-mean diagnostic.

Extension 1: no-predictable-drift diagnostic

A fixed test for artificial conditional mean

Use a simple fixed information set:

$$\omega_t = (1,\ r_t,\ r_{t-1},\ |r_t|,\ |r_{t-1}|)^\top .$$

Define the drift-violation statistic:

$$\delta = \sup_{\|b\|_2 \le 1} \left| \widehat{\mathrm{E}} \left[r_{t+1} b^\top \omega_t \right] \right| .$$

For this linear class:

$$\delta = \left\| \widehat{\mathrm{E}} \left[r_{t+1} \omega_t \right] \right\|_2 .$$

Compare three objects

$$\delta_{\text{real}}, \qquad \delta_{\text{synthetic}}, \qquad \delta_{\text{null}}.$$

The null can be built by block-shuffling returns to preserve rough marginal behavior while breaking predictability.

Extension 2: drift-regularized generation

Keep stylized facts, discourage artificial alpha

Modify training with one additional penalty:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FTS}} + \lambda \delta_{\text{synthetic}}.$$

- $\mathcal{L}_{\text{FTS}}$: original reconstruction, diffusion, and transition losses.
- $\delta_{\text{synthetic}}$: fixed no-predictable-drift statistic.
- λ: chosen once on validation data.

Evaluation question

Does the penalty reduce predictable drift without destroying:

- heavy-tailed returns;
- volatility clustering;
- distributional similarity to real data?

Extension 3: broader model directions

Multivariate FTS-Diffusion

Generate vector returns:

$$r_t \in \mathbb{R}^d$$

and preserve cross-correlations, co-jumps, and sector-level dependence.

Regime-conditioned transitions

Condition the transition module on market state:

$$\phi(p_m, \alpha_m, \beta_m, c_m).$$

Stress-aware generation

Oversample rare but realistic regimes: crises, crashes, liquidity shocks, and volatility spikes.

Stronger validation

Move beyond distribution matching:

- martingale-style diagnostics;
- trading-rule robustness;
- risk metrics such as VaR and ES;
- out-of-sample transfer to new assets.

Takeaway

Main limitation

FTS-Diffusion is strong at generating realistic local motif-based financial paths, but its guarantees are mostly empirical and distributional.

Main extension

Add a fixed no-predictable-drift diagnostic and, optionally, a drift penalty:

$$\text{realistic synthetic data} \quad + \quad \text{no artificial predictability.}$$

This gives a cleaner benchmark for deciding whether synthetic financial time series are useful for augmentation rather than merely visually or marginally realistic.

Conclusion

- FTS-Diffusion addresses the core challenges of irregularity and scale-invariance in financial data.
- SISC allows for clustering recurring market “shapes” effectively.
- The model demonstrates superior performance in preserving statistical properties compared to TimeGAN and QuantGAN.

Outline

Contributions

- Faithful from-scratch replication of FTS-Diffusion (SISC + PGM + PEM) on three assets: S&P 500, GOOG, ZC=F.
- Identified four reproducibility issues in the public reference implementation; documented patches.
- Defined and compared three synthetic-data protocols (authors / split / random_init); showed the headline trend depends on the protocol.
- Added per-run MAPE confidence intervals and a paper-vs.-replication gap diagnostic.
- Pattern dictionaries learned and visualised for $K \in \{11, 13, 14\}$ across all three assets, plus MC transition-state figures.

Open questions

- Which protocol *should* be the benchmark? `split` is the only one that exploits cross-year MC evolution, but stationarity is not guaranteed.
- Is the PEM Markov chain stationary in practice, and on what timescale?
- Why does MAPE grow with augmentation under the authors' protocol: leakage from the rolling-window scheme, or genuine distributional drift?
- How sensitive are downstream results to K? Qualitative flips on ZC=F between $K = 11$ and $K = 13$.

Bottom line
The architecture is sound; the experimental protocol is not. A clean replication requires both.

Backup: repository layout

```
ml-in-fcs/
  architectures/{asset}/k{K}/{data,res,trained_models}/   <- SISC + PGM + PEM ckpts
  synthetic/{asset}/k{K}/{protocol}/run_XX_{syn,states}.npy
  results/tatr/{asset}/k{K}/{protocol}/run_XX.csv          <- per-run MAPE
  figures/{sisc,tatr,mc_analysis}/{asset}/k{K}/...         <- final PDFs
  src/{TATR_Reduced_Replication.ipynb,
       FTS_Diffusion_Training.ipynb,
       B3_SISC_Simulated_Investigation.ipynb}
```

ETH zürich

Deli Lin, Eric Shao, Lapo Linossi, Tom Haidinger

ETH Zurich
Machine Learning in Finance and Complex Systems
Supervisor: Alexander Arandjelovic

Thank you. *Questions?*